

AKASHRAJ KALATHINGAL

Senior Full-Stack Software Engineer | Java • React • Distributed Systems • Cloud Platforms

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 6+ years building high-scale, mission-critical systems across banking, payments, and enterprise data platforms. Proven record designing event-driven microservices in Java/Spring Boot, shipping production React and Angular UIs, and operating distributed workloads on Kubernetes, OpenShift, AWS, and Azure. Currently engineering an LLM-powered financial advisory chatbot at BMO and previously led ISO 20022 modernization for Scotiabank's high-value payment rails. Combines deep systems thinking with a track record of cross-functional leadership, production ownership, and measurable business impact.

PROFESSIONAL EXPERIENCE

Bank of Montreal (BMO) — via Avant Technologies

Apr 2025 – Present

Software Developer, AdviceDirect & AI Advisory Platform

Toronto, ON

- Core engineer on BMO InvestorLine's **AI-driven financial advisory chatbot**, architecting Java/Spring Boot microservices that expose internal banking capabilities as **MCP tools for LLM RAG pipelines**, enabling real-time portfolio insights, buy/sell recommendations, and personalized advice for retail investors.
- Designed and shipped **data ingestion pipelines using Kedro and Azure Document Intelligence** to transform unstructured financial documents into structured embeddings, powering retrieval-augmented generation across the chatbot's knowledge base.
- Led migration of secrets management from OpenShift to **HashiCorp Vault** across 10+ production services, eliminating hardcoded credentials and strengthening compliance posture for regulated banking workloads.
- Drove **on-prem to AWS migration** of critical production services, improving scalability and deployment automation while reducing release lead time through Azure DevOps + Ansible CI/CD pipelines.
- Built modular Angular components and RESTful APIs for the **AdviceDirect onboarding journey**, powering multi-account enrollment flows and improving conversion across self-directed and assisted-investing experiences.
- Deployed and operated microservices on **Red Hat OpenShift** with horizontal autoscaling and zero-downtime rollouts across enterprise Kubernetes clusters.

Scotiabank

Jan 2024 – Feb 2025

Software Engineer — High-Value Payments & Core Banking Platform

Toronto, ON

- Engineered features on Scotiabank's next-generation core payments platform processing **billions of dollars in daily SWIFT and wire transactions**, using Java, Spring Boot, and IBM Financial Transaction Manager (FTM).
- Led **ISO 20022 migration** for outgoing MX wire payments, automating fraud validation, FX conversion, and charges posting — eliminating manual operator steps and aligning Scotiabank with global SWIFT 2025 mandates.
- Owned end-to-end **SSL/TLS certificate remediation** across multiple production payment applications, preventing service disruption and closing audit findings on enterprise security controls.
- Acted as engineering SME on cross-border payment integration, evaluating interrelationships across hubs, fraud engines, and clearing systems to deliver seamless Canadian and international transaction flows.
- Provided production application support on a rotating on-call roster, triaging and resolving incidents on systems handling high-value transaction loads with zero business impact.

Target Corporation

Dec 2019 – Apr 2023

Senior Software Engineer (promoted from Software Engineer) — Enterprise Orchestration Platform Bangalore, India

- Designed and scaled core microservices for Target's **enterprise data orchestration and scheduling platform**, moving petabyte-scale data across heterogeneous systems and powering retail-wide analytics workflows.
- Architected the platform's **Patroller subsystem** for self-healing job recovery, materially improving job success rates and reducing operator pages across peak retail seasons.

- Built the platform's **reporting and analytics layer** (Apache Druid + Grafana) that became the source of truth for leadership KPIs and capacity-planning decisions.
- Drove **platform-wide Log4Shell (CVE-2021-44228) remediation**, coordinating patching across services in a single sprint and hardening dependency hygiene going forward.
- Won the **Target Data Science Hackathon**; recognized with multiple Pyramid Performance Awards and Quarterly Best Performer for sustained delivery and mentorship of junior engineers.
- Modernized the platform UI in **React (Hooks, Context API, Material UI)**, delivering reusable components for scheduling, monitoring, and admin dashboards consumed by hundreds of internal users.

CGI Inc.

Aug 2018 – Dec 2019

Software Engineer

Bangalore, India

- Built an enterprise ERP web application and an event-hosting platform end-to-end, owning backend APIs (Java/Spring), database modeling, and frontend delivery.
- Authored production-grade technical documentation and reproducible runbooks adopted as onboarding material for new hires.

TECHNICAL SKILLS

Languages: Java (8/11/17), TypeScript, JavaScript, Python, SQL, Scala, GraphQL

Backend & Frameworks: Spring Boot, Spring MVC, Spring Batch, REST, gRPC, Kafka, OAuth2/JWT, Microservices, Event-Driven Architecture

Frontend: React (Hooks, Router, Context API), Angular, Material UI, Redux, Cypress

Cloud & Infra: AWS (S3, EKS), Azure, GCP, Kubernetes, Red Hat OpenShift, Docker, Ansible, Terraform-style IaC, HashiCorp Vault

Data & Storage: PostgreSQL, Oracle, IBM DB2, MongoDB, Neo4j, Apache Hadoop, Apache Hive, Apache Spark, Apache Druid

CI/CD & DevOps: Azure DevOps, Jenkins, Drone, GitHub Actions, Maven, Gradle, Git, Bitbucket, SonarQube

Observability: Grafana, Kibana, ELK Stack, Splunk, Filebeat, DOMO

AI / ML: LLM RAG pipelines, MCP (Model Context Protocol), Kedro, Azure Document Intelligence, scikit-learn, pandas, NumPy

Payments & Banking: IBM FTM, IBM IIB, IBM MQ, SWIFT, ISO 20022 (MX), Wire & FX Processing, PCI-aware engineering

EDUCATION

Conestoga College — PG Certificate, Big Data Solutions & Architecture

2023

Waterloo, ON • Graduated with High Distinction; Dean's Honour List (top of class)

UT Austin, McCombs School of Business — PG Program, AI & Machine Learning

2020 – 2021

National Institute of Technology (NIT), Calicut — B.Tech, Computer Science & Engineering

2014 – 2018

AWARDS & RECOGNITION

- Winner — **Target Corporation Data Science Hackathon**
- Pyramid Performance Award (multiple) — Target Corporation & Scotiabank
- Quarterly Best Performer Award — Target Corporation
- Dean's Honour List — Conestoga College (top of cohort)